



ClimaSentinel

Live Climate Risk Dashboard & Automated Climate Pipeline



CLIMASENTINEL

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The Mission / Overview

A live climate dashboard for what is happening now.

ClimaSentinel turns daily weather, air-quality, river and climate-baseline signals into a clear **Global Tipping Risk Score** for 10 major European cities. The core product is not a black-box model: it is a live, explainable dashboard that shows the current risk level, the main threat driver and how each city compares at a glance.



Objectives: Business Impact First

What judges should remember

- Live situational awareness:** a city can be understood in seconds instead of reading multiple weather, pollution and flood feeds.
- One score, many signals:** heat, wind, air quality, river discharge and climate baselines become one simple city-level risk story.
- Decision-ready outputs:** every city receives a risk level, primary threat and plain-language explanation rather than raw technical tables.
- Automated economics:** scheduled cloud ingestion keeps the dashboard lightweight and repeatable.

User experience

- GREEN safe YELLOW watch RED act
- Premium climate-tech interface with color-coded risks, glowing threat indicators and refined progress bars.



Method: Live Climate Intelligence Pipeline

A clean 3-layer climate data factory

1. Collection	BRONZE
Every morning at 06:00 UTC, the system fetches fresh weather, air-quality, flood, ERA5 and CMIP6 data for each city.	
2. Refinement	SILVER
Raw feeds are cleaned, deduplicated and standardized in BigQuery so the dashboard uses trusted city signals.	
3. Live Insights	GOLD
Business tables produce current score, risk color, dominant threat and rankings for the live dashboard.	

Dashboard-first intelligence

- Current climate status:** the main feature is a live view of today’s risk level, main threat and city comparison.
- Experimental forecast:** the 7-day Random Forest module is a research extension tracked with MLflow, not the main product promise.

Raw APIs → Trusted Data → Live Dashboard

Results: Live Dashboard for Climate Now

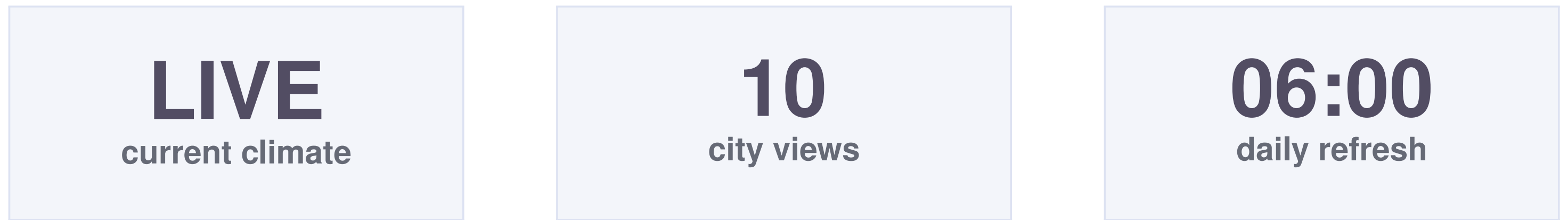
What users understand in seconds

Status Is my city safe? Green / Yellow / Red risk level	Driver Why is risk rising? Heat, air, wind or river pressure	Compare Where is attention needed? 10 cities ranked at a glance
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Dashboard value

- Climate now, not raw data:** judges see the current risk situation instantly across all monitored cities.
- Readable explanations:** every score is paired with the dominant threat so the result feels useful and trustworthy.
- Operational confidence:** automated refresh, validation tests and live monitoring protect the public-facing dashboard.

Prediction note: the 7-day model is a **research module**; the core product is the live dashboard for climate now.



Conclusion: Why ClimaSentinel Matters

From environmental noise to confident action

- ClimaSentinel reframes raw climate feeds as a **live decision product**: understandable, automated and stakeholder-ready.
- The MLOps work makes the dashboard reliable: data ingestion, transformation, versioning, testing, deployment and monitoring all support the live experience.
- The forecast remains an **experimental extension**; the proven value is climate status users can understand today.

“Know the climate risk now; explore what may come next.”

Perspectives: Next Evolution

Roadmap for a stronger live product

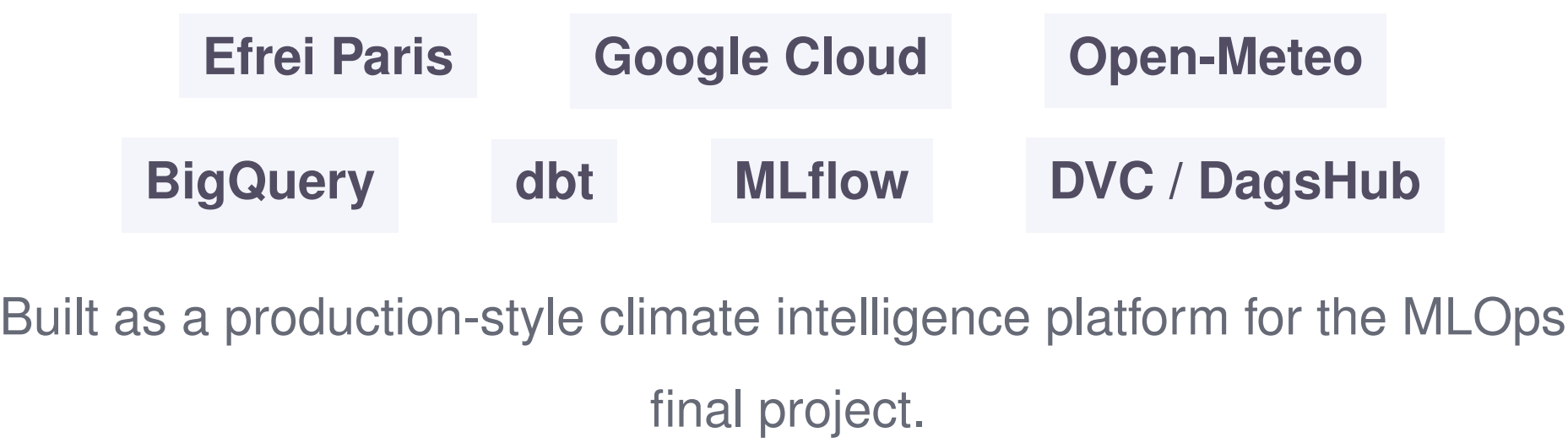
Scale coverage Move from 10 cities to more regions, districts and vulnerable local zones.	Send alerts Email, SMS or municipal webhooks when a risk threshold is crossed.
Explain the score Plain-language drivers, confidence wording and clear city summaries for non-technical users.	Validate forecasts Keep the 7-day model experimental until accuracy, drift and safety gates prove it reliable.

Product principle: publish what is reliable now, label predictions honestly, and make every climate signal easy to act on.

KEY REFERENCES

- [1] Open-Meteo: weather, air quality, flood, ERA5 and CMIP6 APIs.
- [2] Google Cloud, BigQuery, dbt, FastAPI, Next.js and monitoring stack.
- [3] ClimaSentinel GitHub repository, source code and project documentation.

ACKNOWLEDGEMENT AND PARTNERS



MORE INFORMATION



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